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Motivation

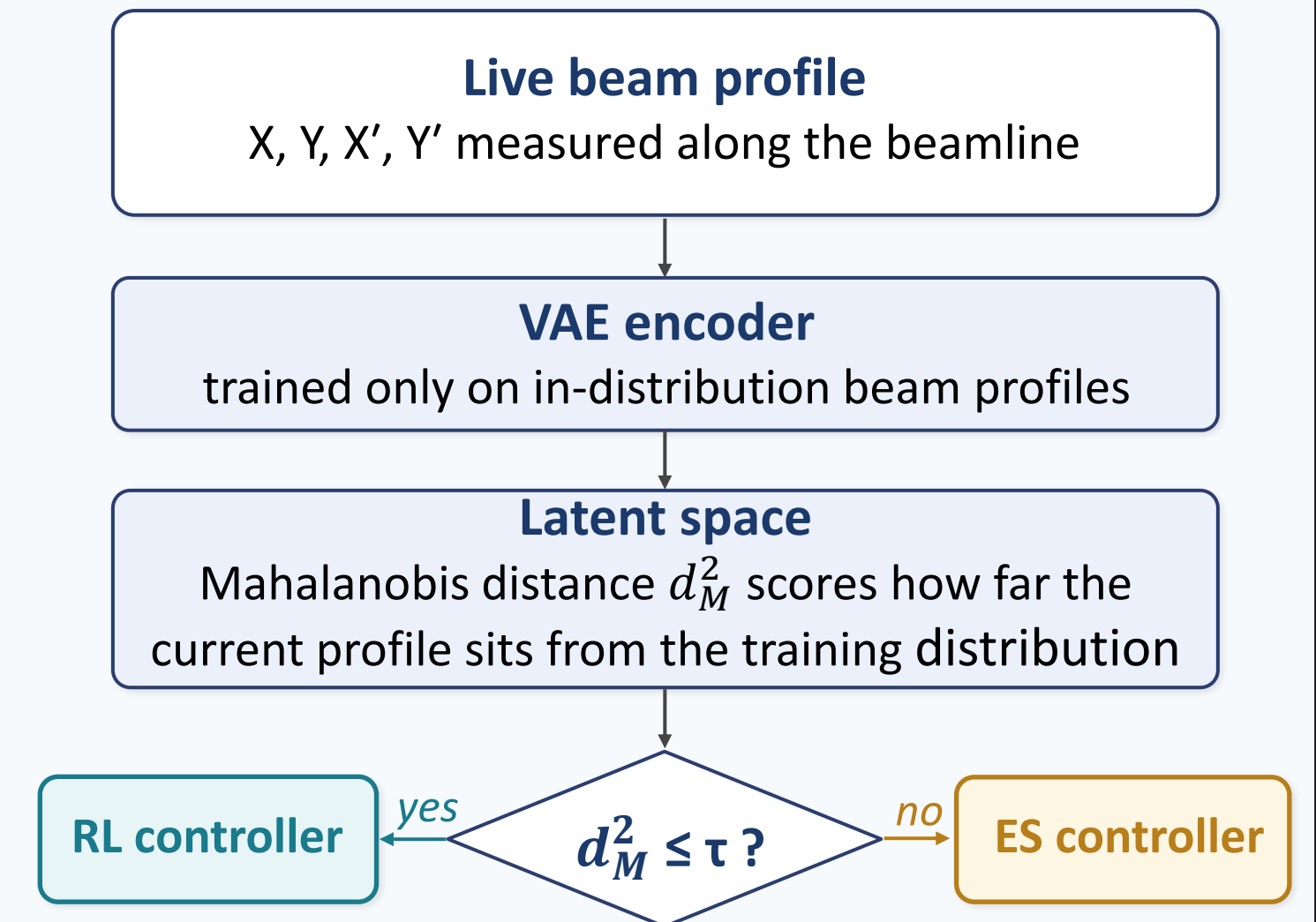
- RL policies trained in fixed environments can fail silently when deployment conditions shift..
- In particle accelerators, magnet drift and lattice changes can push beam profiles out-of-distribution.

How can a controller detect OOD conditions at test time?

Key Idea:

- Learn a VAE latent model of in-distribution beam profiles.
- Score each test-time profile by its Mahalanobis distance in the learned latent space.
- Use that distance as an interpretable binary switch β_t to select between two controllers:
 - RL: fast, in-distribution control
 - Extremum Seeking (ES): robust, model-free fallback under drift

The Big Picture



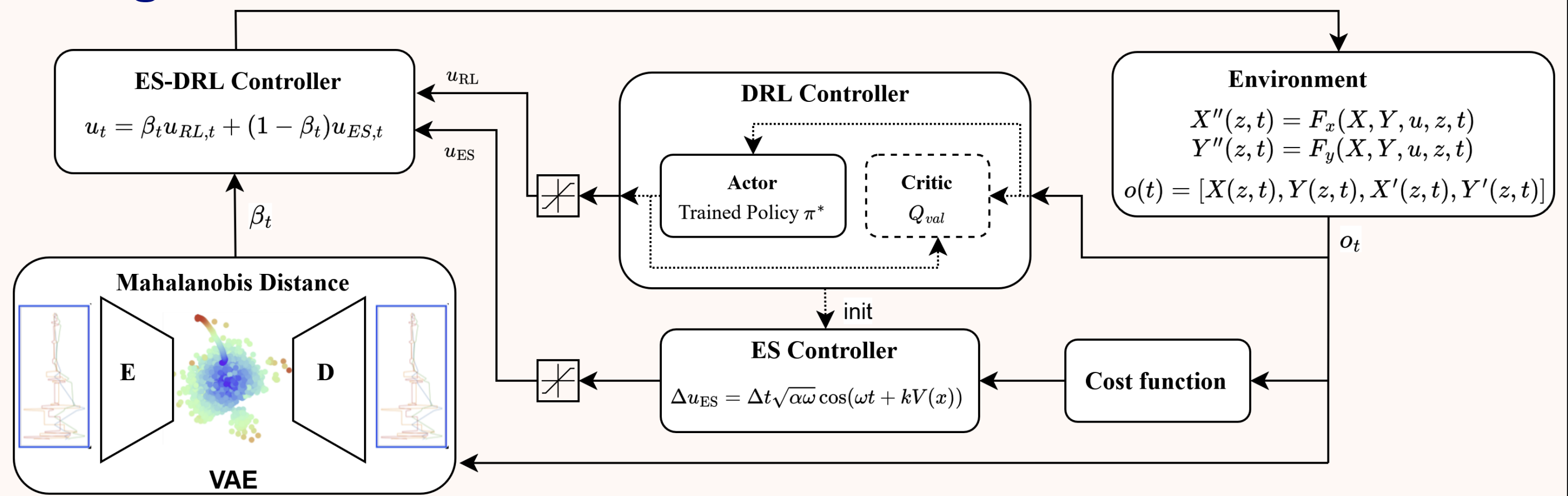
OOD-Aware Test-Time RL-ES Switching

Mahalanobis distance

$$d_M^2 = (z_t - \bar{z})^T \Sigma^{-1} (z_t - \bar{z})$$

z_t : latent embedding of the current beam profile
 $\bar{z}$: mean of in-distribution embeddings
 Σ : covariance of in-distribution embeddings

$$u_t = \beta_t u_{RL} + (1 - \beta_t) u_{ES}, \quad \beta_t = \begin{cases} 1, & d_M^2(x_t) \leq \tau \\ 0, & d_M^2(x_t) > \tau \end{cases}$$

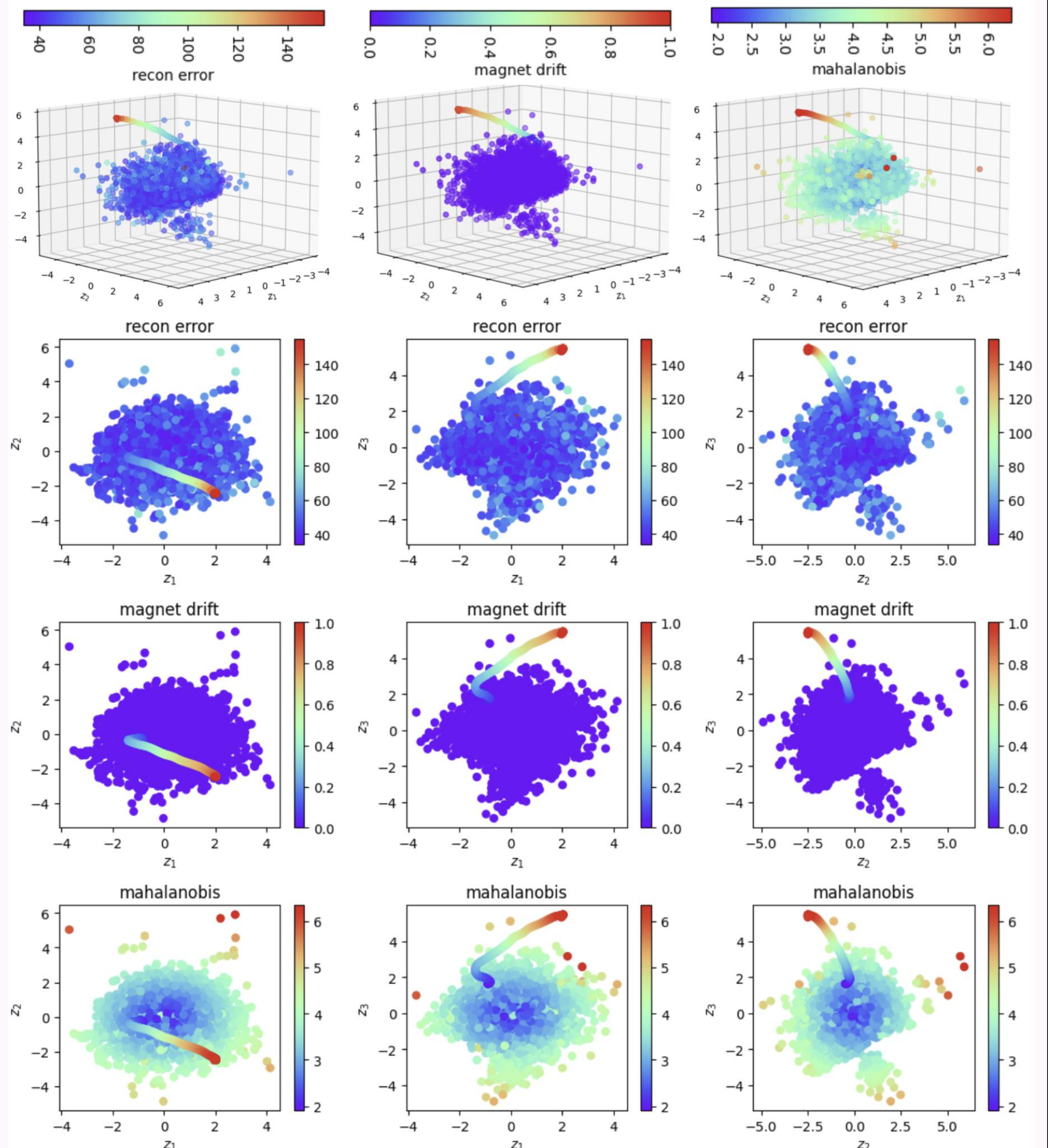
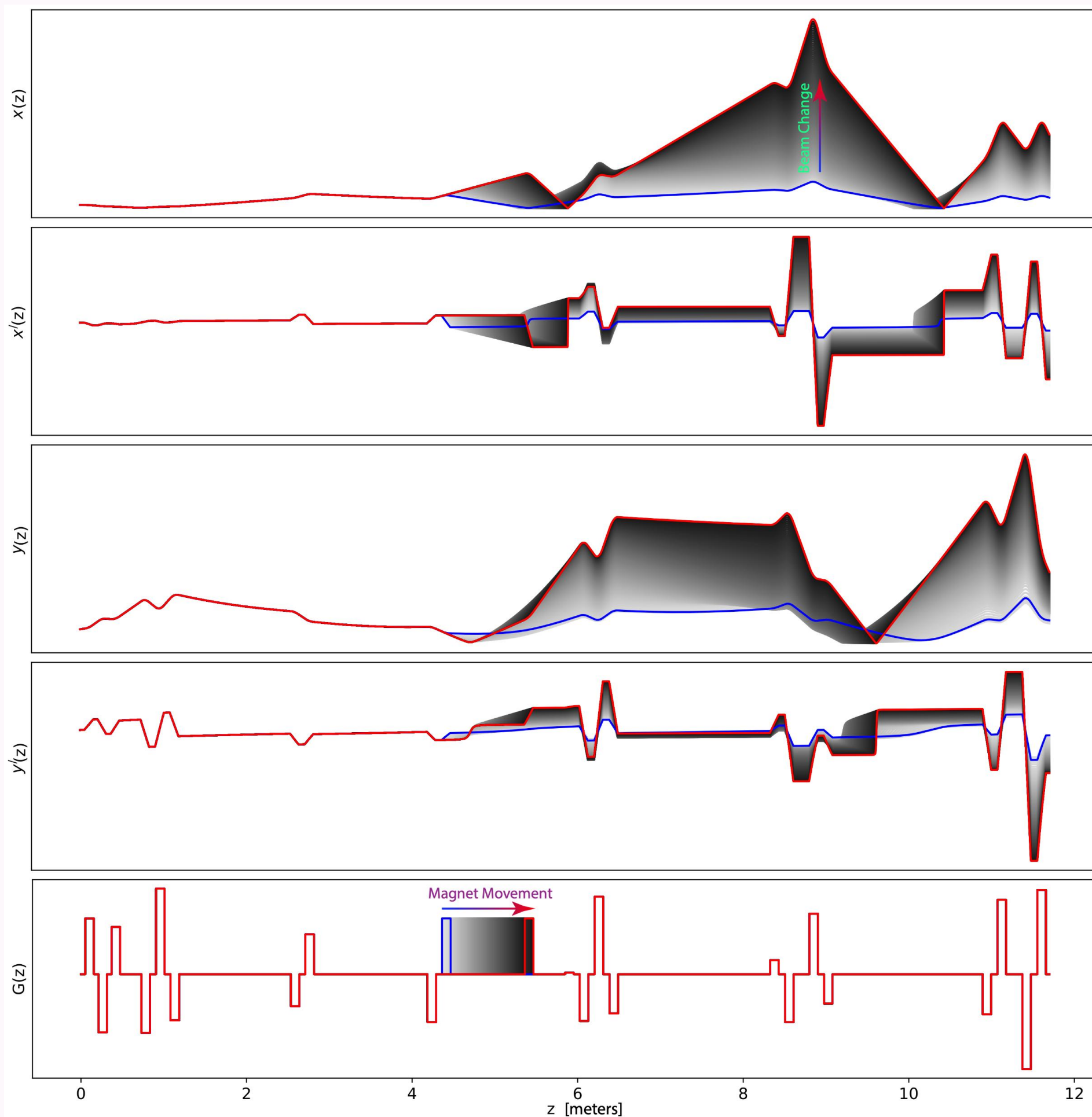


Results: Magnet Drift Creates a Latent OOD Signature

Physical OOD: magnet drift makes beam profiles non-smooth.

Latent signature: VAE embeddings leave the nominal cloud.

Switch trigger: d_M^2 increases with drift, enabling ES fallback.



References:

- Saxena, Shaifalee, Alan Williams, Rafael Fierro, and Alexander Scheinker. "Improved Robustness of Deep Reinforcement Learning for Control of Time-Varying Systems by Bounded Extremum Seeking." *American Control Conference* 2026.
- Saxena, Shaifalee, Rafael Fierro, and Alexander Scheinker. "Deep Reinforcement Learning for Robotic Manipulation under Distribution Shift with Bounded Extremum Seeking." arXiv preprint arXiv:2604.01142 (2026).

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